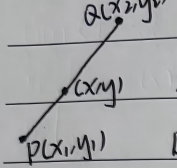
Day 5759 Limits



Why is the equation of PQ: $y-y_1=\frac{y_2-y_1}{x_2-x_1}(x-x_1)$?

Suppose ~~that~~ that there's a moving point (x,y)

Definition of slope: $\frac{y-y_1}{x-x_1}=\frac{y_2-y_1}{x_2-x_1}$ $\therefore y-y_1=\frac{y_2-y_1}{x_2-x_1}(x-x_1)$

$\lim_{x\to c} f(x)=L$

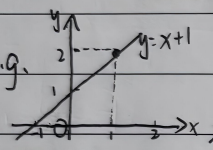
If $f(x)$ is arbitrarily close to the number L for all x sufficiently close to c (other than c itself), we say that f approaches the limit L as x approaches c.

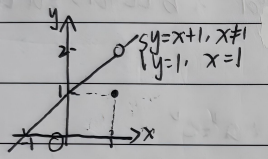
the function value doesn't matter, the trend matters.

$\lim_{x\to c} f(x) \neq f(c)$

$\lim_{n\to\infty}(1+\frac{1}{n})^n=e$

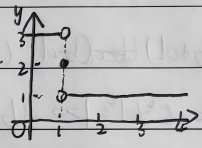
e.g.





both = $\lim_{x\to 1} f(x)=2$

e.g.

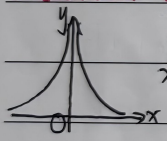


$\lim_{x\to 1^-} f(x)=3$ $\lim_{x\to 1^+} f(x)=1$

$\lim_{x\to 1} f(x)$ does not exist

$\lim_{x\to c} f(x)=L \Leftrightarrow \lim_{x\to c^+} f(x)=L,\ \lim_{x\to c^-} f(x)=L$

unbounded: limit does not exist



$\lim_{x\to 0} f(x)$ does not exist

Day 5760 Limit Laws:

If $\lim_{x\to c} f(x)=L$, $\lim_{x\to c} g(x)=M$, then:

① $\lim_{x\to c}[f(x)\pm g(x)]=L\pm M$

② ~~Lim~~ $\lim_{x\to c}[k\cdot f(x)]=k\cdot L$

③ $\lim_{x\to c}[f(x)\cdot g(x)]=L\cdot M$

④ $\lim_{x\to c}[\frac{f(x)}{g(x)}]=\frac{L}{M}$ $(M\neq 0)$

⑤ $\lim_{x\to c}[f(x)]^n=L^n$

⑥ $\lim_{x\to c}\sqrt[n]{f(x)}=\sqrt[n]{L}=L^{\frac{1}{n}}$

Polynomial Function Limits Law:

$f(x)=a_1x^n+a_2x^{n-1}+a_3x^{n-2}+\dots+a_{n+1}$, $\lim_{x\to c} f(x)=f(c)$

Rational Function Limits Law: $p(x), q(x)$ are polynomials

$f(x)=\frac{p(x)}{q(x)}$ (rational function), $q(x)\neq 0$, then $\lim_{x\to c} f(x)=f(c)$

$\lim_{x\to c} f(g(x)) = f(\lim_{x\to c} g(x))$ ($\lim_{x\to c} g(x)$ exists.) $\lim_{x\to c^+} f(g(x)) = f(\lim_{x\to c^+} g(x))$ $\lim_{x\to c^-} f(g(x)) = f(\lim_{x\to c^-} g(x))$

If $\lim_{x\to c^+} g(x) \neq \lim_{x\to c^-} g(x)$ ($\Rightarrow \lim_{x\to c} g(x)$ does not exist),

$\Downarrow$ $f(\lim_{x\to c^+} g(x)) = f(\lim_{x\to c^-} g(x))$

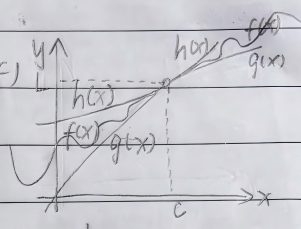
then $\lim_{x\to c} f(g(x)) = \lim_{x\to c^+} f(g(x)) = \lim_{x\to c^-} f(g(x)) = f(\lim_{x\to c^+} g(x)) = f(\lim_{x\to c^-} g(x))$

Day57&6 The Squeeze Theorem:

If $g(x) \le f(x) \le h(x)$ for all x in the interval containing c (except c itself)

$\lim_{x\to c} g(x) = \lim_{x\to c} h(x) = L$,

then $\lim_{x\to c} f(x) = L$



2 Important Trigonometric Limits:

$\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$

$\lim_{\theta\to 0} \frac{1-\cos\theta}{\theta} = 0$

$\lim_{\theta\to 0} \frac{\tan\theta}{\theta} = \lim_{\theta\to 0} \frac{\sin\theta}{\theta} \cdot \frac{1}{\cos\theta} = \lim_{\theta\to 0} \frac{\sin\theta}{\theta} \cdot \lim_{\theta\to 0} \frac{1}{\cos\theta} = 1 \cdot 1 = 1$

The line $y=b$ is a horizontal ~~asympto~~ asymptote of $f(x)$ if $\lim_{x\to+\infty} f(x) = b$ or $\lim_{x\to-\infty} f(x) = b$

The line $x=a$ is a vertical asymptote of $f(x)$ if $\lim_{x\to a^+} = \pm\infty$ or $\lim_{x\to a^-} = \pm\infty$

Day57&8 End behavior model:

the function g is a ~~e~~ right end behavior model if and only if $\lim_{x\to+\infty} \frac{f(x)}{g(x)} = 1$

the function g is a left end behavior model if and only if $\lim_{x\to-\infty} \frac{f(x)}{g(x)} = 1$

} the function g is an end behavior model of f

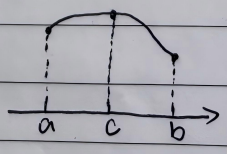
Continuity:

a, b, c is ~~a~~ interior points ~~of~~ in the domain of $f(x)$,

$\lim_{x\to c} f(x) = f(c) \Rightarrow f(x)$ is continuous at c

$\lim_{x\to a^+} f(x) = f(a) \Rightarrow f(x)$ is continuous at left endpoint a

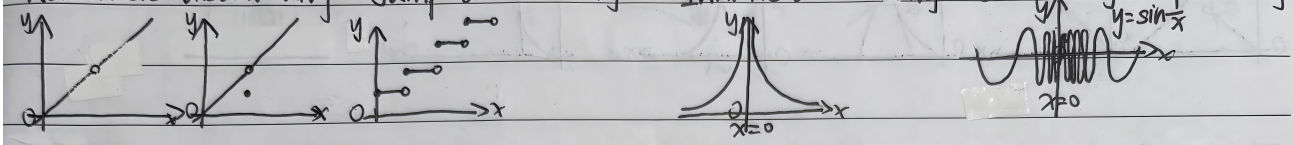
$\lim_{x\to b^-} f(x) = f(b) \Rightarrow f(x)$ is continuous at right endpoint b



Discontinuity = not continuous

4 types of discontinuity

Removable discontinuity | Jump discontinuity | Infinite discontinuity | Oscillating discontinuity

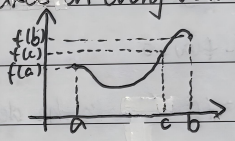


value can be defined/undefined defined value | undefined value | undefined value

IVT: Intermediate Value Theorem for Continuous Functions

A function f that is continuous on a closed interval $[a,b]$ takes on every value between $f(a)$ and $f(b)$

(In other words, we can find some c in $[a,b]$ that $f(a) \le f(c) \le f(b)$



~~Justif~~ Prove: ① f is continuous on the interval $[a,b]$

② $f(a) < f(c) < f(b)$

③ ∴ conclusion

Day 57b Differentiation

The slope of the curve at $x=a$ = the slope of the tangent line to the curve at $x=a$ = $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$

The derivative of a function f at a point x_0 is $f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$ provided ~~the~~ the limit exists.

Differentiation = the process of finding derivatives

Alternate form of the definition of derivative: (at point c) $\lim_{x\to c} \frac{f(x)-f(c)}{x-c}$ provided the limit exists.

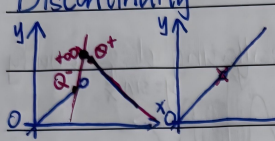
The ~~the~~ derivative of a function is a function f': $f' = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

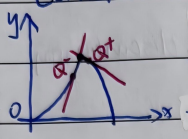
The domain of f' may be smaller than the domain of f

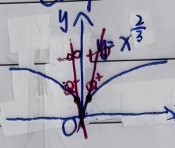
If $f'(x_0)$ exists, we say that f is differentiable at point $x=x_0$

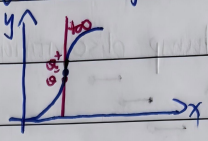
A function is differentiable if every point on its domain is differentiable

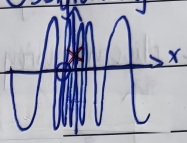
5 Types of functions without a derivative at a point:

Discontinuity 

Corner 

Cusp 

Vertical tangent line 

Oscillating 

If f has a derivative (differentiable) at a point a, then it's continuous at a.

Prove: if f has a derivative at a, $f'(a)=\lim_{x\to a}\frac{f(x)-f(a)}{x-a}$ exists.

$\therefore \lim_{x\to a}[f(x)-f(a)]=\lim_{x\to a}\frac{f(x)-f(a)}{x-a}\cdot\lim_{x\to a}(x-a)=f'(a)\cdot 0=0$

$\therefore \lim_{x\to a}f(x)-\lim_{x\to a}f(a)=0$

$\therefore [\lim_{x\to a}f(x)]-f(a)=0$

$\therefore \lim_{x\to a}f(x)=f(a)$ (that's the definition of continuity)

Day 5780 Differentiation Rule (for Evaluation)

Constant Function: $f(x)=c$ $f'(x)=0$

Power Rule: $f(x)=x^n$ $f'(x)=nx^{n-1}$ $(n\in\mathbb{N}^+)$

→Prove: $\lim_{h\to 0}\frac{(x_0+h)^n-x_0^n}{h}=\lim_{h\to 0}\frac{x_0^n+nx_0^{n-1}h+?x_0^{n-2}h^2+\cdots+nx_0h^{n-1}+h^n-x_0^n}{h}=\lim_{h\to 0}\frac{nx_0^{n-1}h+?x_0^{n-2}h^2+\cdots+h^n}{h}=\lim_{h\to 0}(nx_0^{n-1}+?x_0^{n-2}h+\cdots+h^{n-1})=nx_0^{n-1}$

Constant Multiple: $f(x)=cu(x)\Rightarrow f'(x)=c\cdot u'(x)$ (u is differentiable)

Sum/Difference: $f(x)=u(x)\pm v(x)\Rightarrow f'(x)=u'(x)\pm v'(x)$ (u, v are differentiable)

Product: $f(x)=u(x)v(x)\Rightarrow f'(x)=u'(x)v(x)+u(x)v'(x)$

Quotient: $f(x)=\frac{u(x)}{v(x)}\Rightarrow f'(x)=\frac{u'(x)v(x)-u(x)v'(x)}{v^2(x)}$ $(v(x)\neq 0)$

Day 5787 Derivatives for trigonometric functions

$f(x)=\sin x$ $f'(x)=\cos x$

$f(x)=\cos x$ $f'(x)=-\sin x$

$f(x)=\tan x=\frac{\sin x}{\cos x}$ $f'(x)=\frac{1}{\cos^2 x}=\sec^2 x$

$f(x)=\cot x=\frac{\cos x}{\sin x}$ $f'(x)=-\frac{1}{\sin^2 x}=-\csc^2 x$

$f(x)=\sec x=\frac{1}{\cos x}$ $f'(x)=\frac{\sin x}{\cos^2 x}=\sec x\tan x$

$f(x)=\csc x=\frac{1}{\sin x}$ $f'(x)=-\frac{\cos x}{\sin^2 x}=-\csc x\cot x$

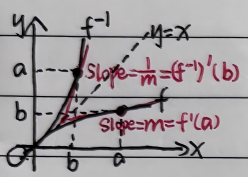
The Chain Rule for Derivatives: If g is differentiable at x, and f is differentiable at $g(x)$,

$(f\circ g)'=f'(g(x))\cdot g'(x)$

$(f\circ g\circ h)'=f'(g(h(x)))\cdot g'(h(x))\cdot h'(x)$

Day 5788 Implicit Differentiation

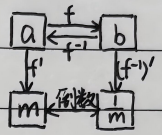
Day 5798 Derivative of Inverse Function



$b=f(a),\ a=f^{-1}(b),\ m=f'(a),\ \frac{1}{m}=(f^{-1})'(b)$

$\Rightarrow (f^{-1})'(b)=\frac{1}{f'(a)}$

$\Rightarrow (f^{-1})'(x_0)=\frac{1}{f'(f^{-1}(x_0))}$



Derivative of Inverse Trig function

① $y=\sin^{-1}x$ domain: $[-1,1]$ range: $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to(\cos\geq 0)$

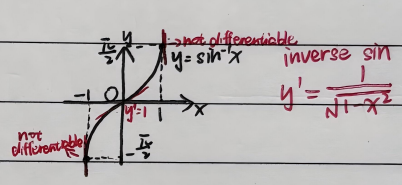
推導: $y=\sin^{-1}x \Rightarrow x=\sin y$

$1=\cos y\cdot y'$

$\frac{dy}{dx}=\frac{1}{\cos y}$ $(\cos y\neq 0\Rightarrow y\neq\pm\frac{\pi}{2}\Rightarrow x\neq\pm 1)$

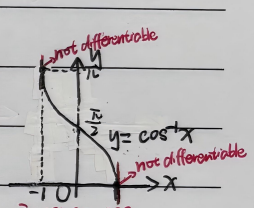
$\because \sin^2 y+\cos^2 y=1 \quad \therefore x^2+\cos^2 y=1 \quad \therefore \cos^2 y=1-x^2$

$\because \cos y\neq 0>0 \quad \therefore \frac{dy}{dx}=\frac{1}{\sqrt{1-x^2}}$



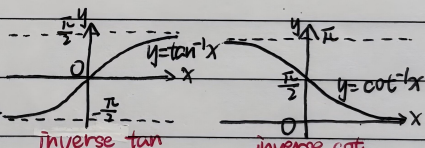
inverse sin

$y'=\frac{1}{\sqrt{1-x^2}}$



inverse cos

$y'=-\frac{1}{\sqrt{1-x^2}}$

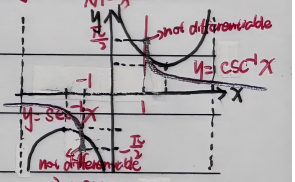


inverse tan

$y'=\frac{1}{1+x^2}$

inverse cot

$y'=-\frac{1}{1+x^2}$

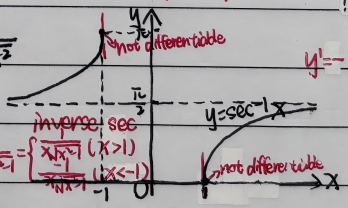


inverse csc

$y'=-\frac{1}{|x|\sqrt{x^2-1}}=\begin{cases}-\frac{1}{x\sqrt{x^2-1}}, & x>1\\ \frac{1}{x\sqrt{x^2-1}}, & x<-1\end{cases}$

② $y=\cos^{-1}x$ domain: $[-1,1]$ range: $[0,\pi]$ $\Rightarrow y'=-\frac{1}{\sqrt{1-x^2}}$

③ $y=\tan^{-1}x$ domain: R range: $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ $\Rightarrow y'=\frac{1}{1+x^2}$



inverse sec

$y'=\frac{1}{|x|\sqrt{x^2-1}}=\begin{cases}\frac{1}{x\sqrt{x^2-1}} & (x>1)\\ -\frac{1}{x\sqrt{x^2-1}} & (x<-1)\end{cases}$

Day 5801 Derivative of Exponential and Logarithmic Functions

$f(x)=e^x \Rightarrow f'(x)=e^x$

$f(x)=a^x \Rightarrow f'(x)=a^x\ln a$ proof: $f(x)=a^x=(e^{\ln a})^x=e^{(\ln a)x} \therefore f'(x)=e^{(\ln a)x}\cdot\ln a=a^x\ln a$ $(a>0,a\neq 1)$

$f(x)=\ln x \Rightarrow f'(x)=\frac{1}{x}$

$f(x)=\log_a x \Rightarrow f'(x)=\frac{1}{x\ln a}$ proof: $f(x)=\log_a x=\frac{\ln x}{\ln a} \therefore f'(x)=\frac{\frac{1}{x}}{\ln a}=\frac{1}{x\ln a}$

e.g. $y=x^x$

$\ln y=\ln x^x=x\ln x$

$\frac{1}{y}\cdot y'=1\cdot\ln x+x\cdot\frac{1}{x}$ $y'=y(\ln x+1)=x^x(\ln x+1)$